Calcitonin. A. Pecile, editor

RATIONALE FOR THE USE OF CALCITONIN IN THE PREVENTION OF POST-MENOPAUSAL OSTEOPOROSIS

TORQUATO NENCIONI, FILIPPO POLVANI
II Clinica Ostetrico-Ginecologica, Università degli Studi, Milano, Italy

INTRODUCTION

Althought postmenopausal osteoporosis occurs so frequently it is still often disregarded in general medical practice, and gynecologists' neglect of this disease state is even more worrying. In our opinion gynecologists should join with internists and orthopedists in studying the prevention of osteoporosis rather than its treatment when clinically manifest and possibly beyond our capabilities to cure.

It has been postulated that the ovary with its endocrine activity plays a major role in the rapid bone resorption which occurs at the onset of the menopause. It seems, therefore appropriate to clarify some aspects of the endocrine picture at the perimenopausal period to ascertain their possible connection with osteoporosis.

THE ENDOCRINOLOGIC PICTURE IN THE PERIMENOPAUSAL WOMAN

The lenght of the menstrual cycle presents notable variations at the two extremes of a woman's reproductive period, the perimenarche and perimenopause. This variable is connected with the occurence of anovular cycles characterized by a lack of progesterone which is secreted in amounts measurable in picograms rather than nanograms as in the reproductive period; in addition , the 17-β-estradiol levels are lower than normal, although this hormone is always present in biologically active concentrations. The lack of progesterone thus precedes cessation of the menstrual flow and determines an endocrinologic picture which has still not been evaluated precisely.

LACK OF PROGESTERONE AND BONE REMODELING

Although progesterone still has no precise place in the treatment and prevention of postmenopausal osteoporosis, there are numerous reports that it may have a direct or indirect effect on bone remodeling in menopausal women. In 1979 Hammond found that the number of factors was reduced in postmenopausal women treated with a combination of estrogens and progestins compared with women treated with estrogens alone. In the same year Johnston reported that the only hormone of those he tested (estrone, estradiol, testosterone, androstenedione and progesterone) able to identify subjects with rapid and slow bone loss at the level of calcium-phosphorus metabolism was progesterone, which was significantly lower in the rapid losers. In 1984 Lobo found a progesterone effect similar to that of the conjugated estrogens, not only with regard to the symptomatology of the climacteric but also in the prevention of bone absorption in postmenopausal women: there was a significant decrease of the calcium/creatinine and hydroxyproline/creatinine ratios both in the group treated with conjugated estrogens and in that treated with high-dose medoxyprogesterone acetate. The mechanism of action of progesterone would be at two levels: 1) it has a high affinity for the corticosteroid receptors, so that it exerts a protective effect; and 2) given in high doses it would thus suppress the adrenal functional up to total suppression of the secretion of glucocorticoids (Wentz, 1977).

We evaluated the loss of bone mass in two groups of subjects by means of densitometric findings using a Norland densitometer with a ^{125}I source. The observation period was 6 months. The first group consisted of 14 menopausal women (mean age 49.7 years, range 45-56) with amenorrhea and elevated FSH values confirming their menopausal status. The second group consisted of 14 premenopausal subjects (mean age 48.3 years, range 46-51) with irregular menstruation and FSH values at the upper limits of normal. The loss of bone mass in the two groups was comparable (fig. 1), indicating that the process of rapid bone resorption starts before the onset of amenorrhea and the abrupt fall in the estrogen levels and coincides with decline in progesterone secretion. This finding seems important with regard to the prevention of postmenopausal osteoporosis and timing the start of possible treatment.

THE ENDOCRINOLOGIC PICTURE AT THE MENOPAUSE

With the onset of amenorrhea the cyclic activity of the ovary ceases definitively. Its function does not cease but undergoes qualitative and quantitative changes. Estrogens are always formed from the androgen metabolites, androstenedione and testosterone, both in women in reproductive age and in menopausal women; androstenedione and testosterone are thus metabolized to estrogens both at the ovary and the periphery (adipose tissue, skin, hair follicles brain,

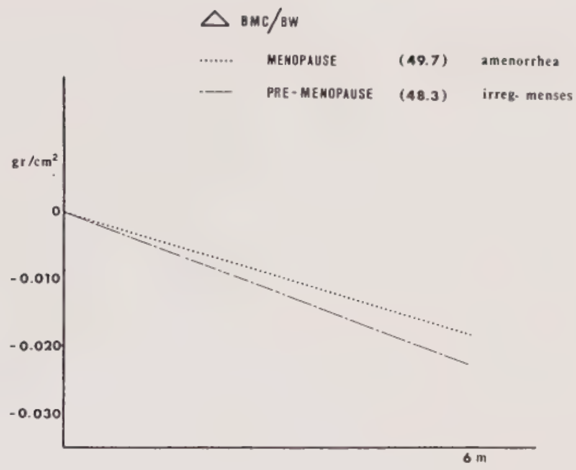

Fig. 1 - Loss of bone mass expressed ad the △ variation of densitometric findings in 14 menopausal and 14 premenopausal women.

bone, muscle) in women reproductive age but only at the periphery in menopausal women. Judd performed a classic study of endocrine activity of the ovary in the postmenopause (1976), and found a reduction of 50% of testosterone levels and of about 20% of androstenedione levels in postmenopausal women after oophorectomy. This indicates surgery of the ovary should be conservative even in women who have passed their normal reproductive period. Despite this endocrinologic findings which orientate gynecologists to prescribing hormone replacement treatment in the menopause, and not forgetting the genital atrophy and psychosomatic phenomena connected with this period of a woman's life, the feedback loop connecting calcitonin to the estrogens is still a valid hypothesis, and is the rational on which is based the use of calcitonin in treatment for the prevention of postmenopausal osteoporosis.

SELECTION CRITERIA FOR SUBJECTS AT RISK OF POSTMENOPAUSAL OSTEOPOROSIS

The criteria for selection of menopausal women who may undergo rapid bone resorption are still far from being clearly defined. Starting, however from data in the recent literature, there is a correlation between the climacteric symptomatology expressed as Kupperman's index (which considers vasomotor disorders, paresthesia, insomnia, irritability, vertigo, arthralgia, headache, palpitations and fatigability) and the lack of estrogen precursors such as androstenedione (Mango, 1984). We investigated if the climacteric symptomatology expressed with this index is related to basal densitometric findings in a study of 100 women (mean age 51 years, range 48-54) with amenorrhea and high FSH levels in whom climacteric symptomatology had appeared not more than 12 months previously. The correlation between the above two parameters was significant ($p < 0.05$; fig. 2). This finding must be carefully assessed, and the method may have a certain use in selecting subject at risk, since it is simple to perform. Already in 1981 Christensen observed a significant correlation between densitometric findings and height, weight and line body mass in menopausal subjects.

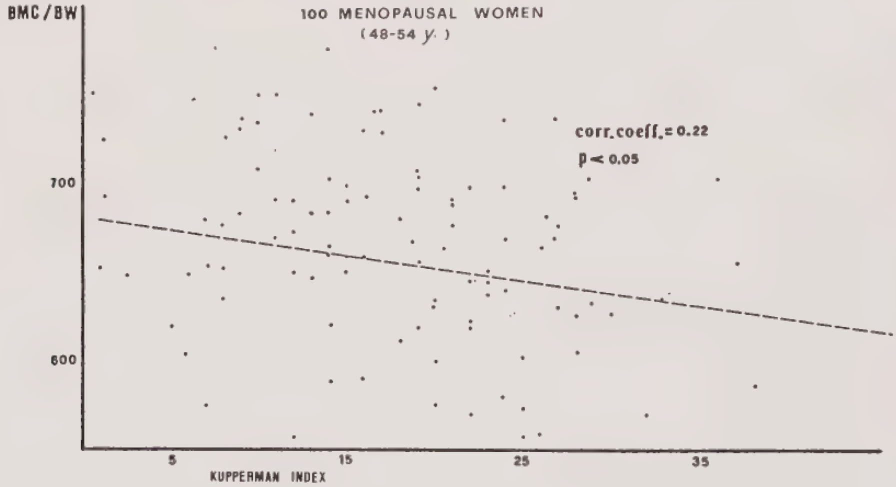

Fig. 2 - Correlation between bone densitometric findings (BMC/BW) and Kupperman's index.

SALMON CALCITONIN

We are performing a study on long-term treatment with salmon calcitonin in a group of 14 menopausal women (mean age 50.6 years) with densitometric values below the normal range. Table 1 shows the treatment schedule. The control group (mean age 50.3 years) consists of 14 menopausal subjects with densitometric values in the normal range. The subjects in the treated group are given 50 MRC units of salmon calcitonin on alternate days and alternate months, plus calcium (1g/day). The planned duration of the study is 12 months; the preliminary results at 6 months are presented. Follow-up includes determinations of the parameters listed in the lower part of Table 1.

EXPERIMENTAL DESIGN

° TWO GROUPS OF AGE-MATCHED PATIENS:

Group 1 = 14 postmenopausal patients with normal densitometric values (control group)

Group 2 = 14 postmenopausal patients with low densitometric values (treated group)

° TREATMENT

50 MRC units of salmon calcitonin of alternate days and alternate months + calcium 1g/day

Duration of treatment = 12 months

° METHODS

Pain intensity
Bone mass densitometry
Plasma alkaline phosphatase
Serum calcium and phosphorus

Tab. 1 - Experimental design for study of salmon calcitonin

Calcemia at 6 months' treatment was reduced significantly ($p<0.05$); also the fall of alkaline phosphatase levels was highly significant ($p < 0.001$).

Fig. 3 shows the $\triangle$ variation of the densitometric values at 6 months in the control and in the calcitonin-treated groups; as can be seen, although there is a certain data scattering, the control group had a bone loss of about 0.020 g/m^2 whereas the basal values were maintained in the treated subjects. The difference between the densitometric values in the two groups, expressed as the $\triangle$ variation compared with the basal value, is statistically significant ($p < 0.05$).

CARBOCALCITONIN

We also started another study at the same time, administering carbocalcitonin at three different doses to menopausal women with densitometric values below the normal range. The experimental design (table 2) includes comparisons between the three groups of subjects treated with different doses of carbocalcitonin, and between these three groups considered together and a control group of untreated

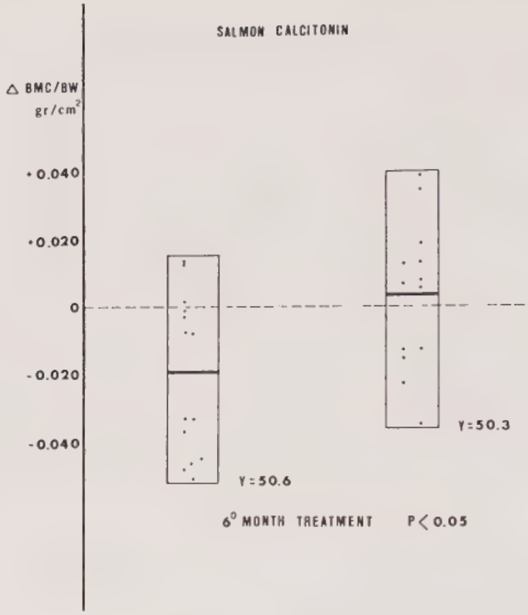

Fig. 3 - The △ variation of the bone densitometric finding after 6 months in the control group (on the left) and the group treated with salmon calcitonin (on the right).

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CONTROLLED TRIAL ON THE EFFECTS OF THREE DIFFERENT DOSES OF CARBOCALCITONIN IN POSTMENOPAUSAL OSTEOPENIC PATIENTS

° <u>Experimental design</u>

- between - patients comparison
- n = 15 patients per dosage group
- n = 15 patients as untreated control group
- doses: 10 - 20 - 40 MRCu 3 times/week + calcium 1g/day
- duration of treatment: 12 months

° <u>Methods</u>

- pain intensity
- bone mass densitometry
- urinary hydroxyproline
- plasma alkaline phosphatase
- serum calcium and phosphorus
- urinary calcium and phosphorus
- side effects (check-list)

Tab. 2 - Experimental design for study carbocalcitonin age-matched menopausal subjects with normal densitometric values. dosage of carbocalcitonin is 10, 20 and 40 MRC units, 3 times a week, plus calcium (1g/day). The planned duration of the study is 12 months. The tests to be performed on the subjects are shown in the lower part of table 2. Spinal pain on movement has decreased significantly in all three treated groups (fig. 4).

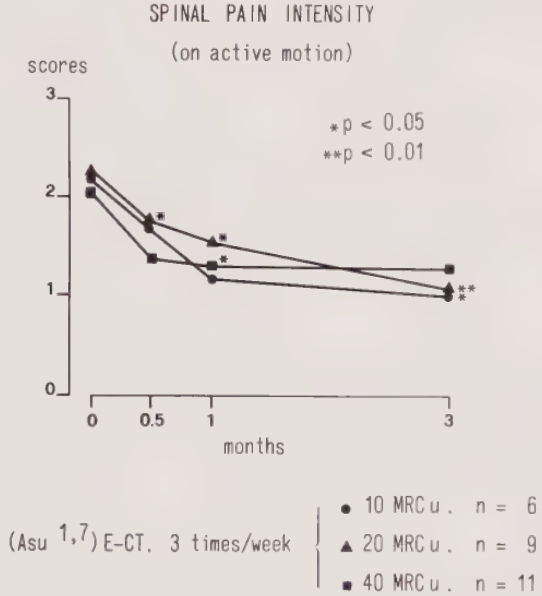

Fig. 4 - Analgesic effect of carbocalcitonin at three different doses.

The analgesic effect in the group treated with 20 units was already evident at 15 days, whereas in the other groups it was observed later; however, not all the women being sudied are included in this and the following results since the study as still not been completed. Calcemia in 27 of the treated subjects from all the groups showed a decrease which was significant at 3 months and then reached a plateau at 6 months. Likewise the alkaline phosphatase pattern showed a highly significant decrease considering the treated subjects together, whereas the variations are attenuated if the groups are considered separately, although still at the limits of statistical significance. It is interesting that the pattern of hydroxyproline secretion correlated inversely with the △ increase of the densitometric value, and that the group of women treated with the highest dose of carbocalcitonin showed the greatest degrease in hydroxyprolinuria, with at the same time the greatest increase in the densitometric finding. In fig. 5 the treatment duration in the four groups studied is shown in the abscissa, and the △ variation of the densitometric finding in the ordinate.

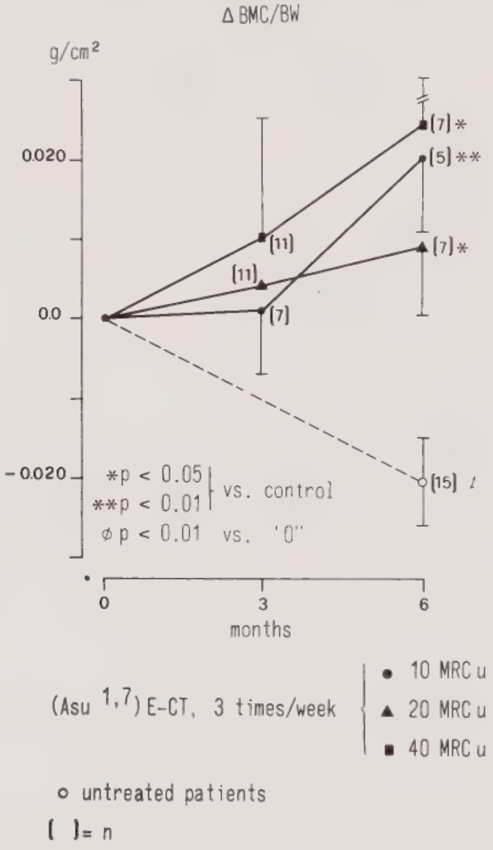

Fig. 5 - The △ variation of bone densitometric findings after 6 months in the central group (dotted line) and in the 3 groups treated with carbocalcitonin (number of subject in parentheses)

The decrease of the densitometric finding in the control group (dotted line) is highly significant compared with the basal value. The △ increase in the subjects treated with carbocalcitonin was significant compared with the finding in the control group, although the number of women who have been treated for 6 months is still small since the study has not yet been completed. However, the data obtained up to now seem to support the administration of calcitonin to menopausal women for the prevention rather than the treatment of postmenopausal osteoporosis. To achieve maximum prevention, treatment should start before the onset of the menopause, and should theoretically be continued during the years in which menopausal women demonstrate the greatest lost of bone tissue. The administration route of the drug must now be considered, since the parenteral route is without doubt the least acceptable to the women concerned, expecially when the treatment is one of prevention.

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